



ALBUKHARY INTERNATIONAL UNIVERSITY



ASSIGNMENT 2:
Password Security:
Safeguarding Digital Identities in the AIU Community

CCS2263- CYBERSECURITY SYSTEMS

Name:

OMID QAZIKHIL

Students' ID

AIU22102314

GROUP:

F

2024-2025
DR NOR HANIZA BINTI MOHD KHIR
norhaniza.khir@aiu.edu.my

Password Security: Safeguarding Digital Identities in the AIU Community

1. Introduction

To avoid falling victim to cyberattacks, cybersecurity should be implemented by everyone, including companies. As more work, communication, and learning are now conducted online, AIU emphasizes the importance of possessing strong cybersecurity knowledge. The primary focus of this paper is password protection, which is fundamental yet often overlooked in cybersecurity.

The primary method for preventing unauthorized access to online resources is the use of passwords. Having a weak or stolen password makes you vulnerable to identity theft, data handling errors, and financial loss. All members of the AIU group, including students, staff, and faculty, must appreciate the importance of strong passwords and two-factor authentication (2FA) to keep the community safe(Grassi et al., 2017).

2. Background Information

A password security plan consists of methods that are used to develop, handle, and protect passwords from being stolen and misused. Picking passwords that are secure and special is the main thing that matters...

The need to protect against more advanced cyber hackers has caused people to adopt alphanumeric strings instead of regular words or numbers as passwords. Since early computer systems only accepted simple passwords, they were easily accessed by hackers, as people commonly ran multiple accounts with the same password. As a result of numerous brute force attacks, dictionary attacks, and credential stuffing, it is necessary to implement stronger password laws and authentication methods.

Research conducted in 2024 has shown that over 80% of hacking-based security breaches involve the use of a compromised password. To add, around 59% of people continue to use one password for different websites, exposing them to greater risk, the poll revealed in 2023(*Verizon Business*, n.d.).

3. Impact on Individuals and Organizations

Both individuals and companies are endangered when people use weak passwords. Access to an AIU student's account by a stranger might result in them accessing such documents as financial, personal email, and academic records. Two common effects of such identity theft are credit fraud and adverse effects on a person's reputation(Federal Bureau of Investigation, n.d.).

If an attacker gains access to passwords, it can compromise the overall security of the institution's critical systems. Obtaining university data without permission could result in private information becoming accessible, which may include research results, employee compensation, and students' test grades This kind of incident may harm AIU, result in fines, and necessitate expensive repairs to rectify.

For instance, a major university experienced a breach in 2019 when unknown attackers compromised thousands of accounts by exploiting weak passwords, resulting in large-scale identity theft and significant issues for the university(Wikipedia contributors, 2025).. This raises the point that we should ideally manage our passwords in advance and remind people to watch out.

4. Preventive Measures and Best Practices

To mitigate risks associated with password vulnerabilities, individuals and institutions must adopt comprehensive security protocols.

For Individuals:

- Use strong passwords combining uppercase and lowercase letters, numbers, and special characters. Avoid easily guessable information such as birthdays or common words(NIST, 2020).
- Employ unique passwords for different accounts to prevent credential stuffing attacks.
- Enable two-factor authentication (2FA) whenever possible, adding an extra layer of security by requiring a secondary verification method like a code sent to a mobile device(Google, 2019).
- Regularly update passwords and avoid reusing old passwords.
- Utilize password managers to securely store and generate complex passwords without the need to memorize them.

For AIU as an Institution:

- Implement a mandatory password policy requiring complexity, expiration, and uniqueness across all university systems.
- Enforce 2FA on all critical platforms including student portals, email accounts, and administrative tools.
- Conduct regular cybersecurity training workshops to educate the AIU community about password best practices and emerging threats.
- Deploy technical solutions like account lockouts after multiple failed login attempts to deter brute force attacks.
- Provide easily accessible resources, including guides and tutorials, on password security.

By applying these measures, AIU can significantly reduce its vulnerability to cyberattacks and protect both institutional assets and personal data of its community members.

5. Conclusion

The topic of password security as it protects digital identities among AIU members was highlighted in this paper. Apart from putting each individual at risk, weak passwords and poor password management put the university at risk of significant financial losses and damage to its reputation. Secure cybersecurity can be achieved by ensuring that everyone creates strong passwords, uses two-factor authentication, and makes password standards a top priority. Providing excellent password protection advice and regularly updating security, including the adoption of new technology tools, is essential. All staff and students should also take steps to protect their accounts and utilize available security resources. Everyone in the AIU community must take part in cybersecurity to support a safer online environment.

References

- Grassi, P. A., Fenton, J. L., Newton, E. M., Perlner, R. A., Regenscheid, A. R., Burr, W. E., Richer, J. P., Privacy Authors, & Usability Authors. (2017). NIST Special Publication 800-63B Digital Identity Guidelines: Authentication and Lifecycle Management. In *NIST Special Publication 800-63B* [Report].
<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-63b.pdf>
- Verizon Business. (n.d.). Verizon Business.
<https://www.verizon.com/business/resources/reports/dbir/>
- Federal Bureau of Investigation. (n.d.). *2021 Internet Crime Report*.
https://www.ic3.gov/Media/PDF/AnnualReport/2021_IC3Report.pdf
- Wikipedia contributors. (2025, May 29). *List of data breaches*. Wikipedia.
https://en.wikipedia.org/wiki/List_of_data_breaches
- Google. (2019, May 17). *New research: How effective is basic account hygiene at preventing hijacking*. Google Online Security Blog.
<https://security.googleblog.com/2019/05/new-research-how-effective-is-basic.html>
- National Institute of Standards and Technology (NIST). (2020). *Digital Identity Guidelines* (NIST SP 800-63B).
<https://doi.org/10.6028/NIST.SP.800-63b>